

Point-by-point response to R1

The reviewer's original comment is reproduced under **Reviewer comment**. Our response, new evidence, and revision action follow under separate labels so that the editorial record is auditable.

R1-benefits

Reviewer comment This article used two locked operators to a training-free composition estimator. The benefits are: a per-spot malignant fraction is written only where the reference can separate the malignant program from its neighbors: a KEEP/ABSTAIN reportability gate that withholds the malignant coordinate once the malignant-neighbor cosine reaches a frozen cutoff. A Patient ID donor split builds mixed spots and reference from disjoint patients. The four-patient CosMx basal-cell carcinoma experiment recovers named tissue objects. The frozen cosine reports KEEP, and therefore a tumor fraction; it returns ABSTAIN, holding the malignant coordinate missing and continuing the remaining types on the simplex. The cutoff separates collinear type pairs rather than assay platforms.

Authors' response and evidence

Comment. KEEP/ABSTAIN and Patient-ID disjoint donor split are benefits.

Response. We thank the reviewer for identifying the two operational contributions. We now separate the fixed baseline implementation from the public SPGD archive from the two operators evaluated here. The gate is a reportability state: when the malignant signature is too collinear with an eligible neighbor at the frozen cutoff, the malignant coordinate is withheld rather than silently reported. The donor split is a patient-level separation between mixed-spot construction and reference construction. We preserve the historical locked outputs and add an explicit algorithm panel, frozen-decision inventory, and reproducibility instructions so that the benefit can be inspected without inferring it from prose.

New evidence. The contribution is supported by the frozen $c^*=0.80$ gate, explicit missing malignant coordinate, and patient-disjoint donor construction; no new biological specimen is needed.

Revision action completed. The contribution-to-algorithm map, fail-closed state diagram, frozen-constant inventory, and donor-disjoint construction are now included in the revised Methods and protocol figure; the expanded results evidence is cross-referenced in F13–F14.

R1-usability

Reviewer comment Disadvantage is: (1) require more practice; (2) require more lab support.

Authors' response and evidence

Comment. The method may require more practice and laboratory support.

Response. We interpret the practice and laboratory-support comment as a request that the computational protocol be usable by an independent reader, not as evidence that the method requires wet-lab support. We therefore added a reader-facing quick-start path, tested environment and hardware information, repeated runtime summaries, explicit refusal behavior, and troubleshooting for missing or incompatible inputs. The archive audit confirms that the

locked-result and rendering workflow is reproducible from the released plot data and documented inputs. A complete raw-data-to-fit rebuild still depends on external reference-building inputs and licensed raw libraries, which are stated as a boundary rather than presented as silently available.

New evidence. The archive is reproducible at the locked-result and figure-rendering level: the public source revision, frozen decisions, donor-disjoint status, plot data, rendering material, and test contract agree. The audit also records the boundary between this verification capsule and a complete raw-data-to-fit reconstruction, which requires external and licensed inputs.

Revision action completed. The archive and response package now provide a reader-facing quick-start path, environment/runtime record, troubleshooting boundary, and explicit computational/no-new-specimen statement. The remaining raw-data-to-fit limitation is stated directly.

Reproducibility boundary

The package is therefore described as a locked-analysis and render capsule. It supports verification of frozen constants, donor-disjoint status, plotdata, rendered figures, version-consistency checks, and the recorded test contract. It does not claim that a reader can reconstruct licensed raw data or external reference-building inputs without access to those sources. This distinction is now stated in the Methods, Discussion, Data Availability text, and supplementary README.